

When Does Admission Control Pay? Decomposing End-to-End Wall Clock in Memory-Bound Agentic LLM Serving

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Abstract—Agentic LLM workloads differ from chat in a way that makes admission control look like the obvious lever: each session is a long, multi-round job whose KV footprint grows every round, and a single session can occupy most of a memory-constrained GPU’s KV pool. A natural hypothesis follows—a controller that reads engine telemetry and admits on measured capacity should beat a static concurrency cap. We test that hypothesis directly and find that, on our workload and hardware, it is false: across nine controller variants, none lands above the static Pareto front, and a plain cap on resident sessions remains the fastest policy we measured.

We explain why, and the explanation is the contribution. We decompose the end-to-end wall clock into *busy* and *idle* time, and show that the busy component is governed by two quantities the admission decision does not control: the per-step cost and the number of requests actually in flight. Admission policy moves the second—but the first rises with it, so the two cancel. The idle component (7.4–36.3% of the wall clock) is structural: an agent session issues no request while waiting on a tool, and we measure that this wait averages 16.85 s, not the sub-second figure a naive reading of the trace suggests. We further show that the same throttle setting that is worth 51% at low load costs 42% at high load—so no single scalar can be optimal across regimes, which is precisely the regime in which adaptive control could have paid. We report four mechanistically distinct policy routes we falsified, each with its measured reason, and we argue that prior systems’ reported gains from state-preserving suspension are already realised by the permit-per-session design our controller inherits—which is why adding adaptivity on top of it yields nothing.

Index Terms—LLM serving, admission control, KV cache, agentic workloads, performance characterization, negative results

I. INTRODUCTION

An agentic LLM session is not a request. It is a long-lived job that issues a round of inference, waits on a tool, and comes back—tens of times—with a context that grows every round. On a memory-constrained accelerator this makes admission control look like the obvious lever. The KV pool is the binding resource; a single large session can occupy most of it; and the sessions themselves are wildly heterogeneous. If a scheduler can see how much of the pool a candidate will need and how much room is left, surely it should beat a fixed number.

We believed this, built a controller that does exactly that, and spent a considerable amount of GPU time discovering that on our workload and hardware it is false. This paper reports the negative result, but its contribution is not the negative result—it is the decomposition that explains it, and the boundary that decomposition draws.

A. What we measured

We serve a replay of 200 real coding-agent sessions on a single RTX 3090 (Qwen2.5-Coder-7B, 96K context, SGLang, a 101,432-token KV pool) and compare a family of admission policies against a sweep of static concurrency caps. Nine controller variants, spanning four mechanistically distinct ideas, all land on or below the static Pareto front (Fig. 4). The best of them reaches the front and no further.

B. Why

We decompose the wall clock into *busy* time, during which the engine has at least one request in flight, and *idle* time, during which it has none. The busy component is governed by

$$W_{\text{busy}} \approx \frac{\text{output tokens}}{\bar{R}} \times \tau, \quad (1)$$

where $\bar{R}$ is the mean number of running requests and τ is the per-step cost. An admission policy controls $\bar{R}$ and nothing else. The difficulty is that τ is not constant in $\bar{R}$: we observe it rise by 20–25% in exactly the two configurations that push concurrency hardest, and the rise cancels the gain from the extra concurrency.

The idle component is structural rather than a control failure. A session in a tool wait issues no request, so when all live sessions happen to be waiting, the engine is genuinely empty—and we measure that stretches of 5–25 s of zero occupancy are common. Crucially, the tool wait itself is much longer than the trace’s median suggests: the median inter-round gap is 0.35 s, but the *mean* is 16.85 s with 22% of gaps over ten seconds. A controller that reads only the median concludes, as we initially did, that sessions never leave and therefore that there is no idle time to recover. There is.

C. Contributions

- We falsify, with pre-registered criteria, four mechanistically distinct routes by which an adaptive admission controller was expected to beat a tuned static cap on this workload: changing the accounting base, changing the growth reserve, changing the hard ceiling, and changing the execution granularity from session-count to running-request-count. Each is reported with its measured reason (§IV).
- We give a two-part decomposition of the wall clock (§III) that explains all of the above with one mechanism:

admission policy moves the running-request count, and the per-step cost moves with it.

- We characterize when adaptivity *could* have paid, and show that our workload is not that case: the same throttle configuration that is worth +51% at 25 sessions costs −42% at 200 (§V). Put differently, no single scalar cap is optimal across both, which is the regime adaptive control exists for—and we measure that adaptive control does not reach it either, because the aggregator’s own accounting error is load-independent.
- We make the measurements reproducible: the verdict of every run is produced by a single script from the raw per-step logs, and we report the three claims we had to retract along the way, with the checks that caught them.

D. What we do not claim

We do not claim that admission control is useless in general. We claim it does not pay *here*, we show the measurement that decides it, and we argue (§VII) that the same measurement can be taken on another workload in minutes. We also do not claim our hardware and model are representative; our results are from one 24 GB accelerator and one 7B model, and extending them is future work.

II. BACKGROUND AND SETUP

A. Workload

We replay 200 coding-agent sessions drawn from real traces. Each session is a sequence of rounds; a round presents a prompt of median 23,187 tokens (p90 51,064; maximum 89,436, which alone is 88% of the KV pool), and between rounds the session waits on a tool before returning with a longer context. Arrivals are Poisson at $\lambda = 0.04 \text{ s}^{-1}$ over an 80-minute window, giving 999 steps in total. Two sessions of maximum size cannot be co-resident: their combined contexts exceed the pool by 76%.

The tool wait is the load’s defining property and the one most easily misread. Its *median* is 0.35 s, which invites the conclusion that sessions never leave the engine. Its *mean* is 16.85 s, 22% of waits exceed ten seconds, and the maximum is 300 s. We verified that our replay harness honors the trace faithfully: across 799 round transitions the measured gap is 16.85 s mean against a demanded 15.82 s, a ratio of 1.001. This matches the source corpus’s own reported mean tool time of 16.8 s.

B. Engine and the binding constraint

We use SGLang v0.5.10 with Qwen2.5-Coder-7B-Instruct extended to 96K context, on one RTX 3090 (24 GB). At `mem-fraction-static 0.85` the KV pool is 101,432 tokens. Hierarchical caching (HiCache) is enabled throughout, with a host tier of twice the device pool. Every run is guarded: if the pool does not come out at $101,432 \pm 20$ —which happens when another tenant holds memory on the card—the run is aborted rather than recorded, because SGLang sizes the pool from free memory at startup and a shrunken pool silently produces incomparable numbers.

The relevant consequence is that the pool holds only about four median sessions. Concurrency is therefore capacity-bound, not scheduler-bound, and this is the premise on which everything below rests.

C. The controller under test

Foyer is an admission controller that reads engine telemetry and issues *named* permits rather than a scalar cap: a session is admitted by appearing in a permit file, and the runner refuses to start a round for a session that is not listed. This design choice matters for the comparison, because it is what allows per-session actions at all—including the pause primitive we exploit in §IV-E. It also means that a session, once admitted, holds its permit for its entire lifetime and its KV is preserved across tool waits. We return to this point in §VII, because we believe it is why published gains from state-preserving suspension do not reappear here.

Each control cycle (2 s) the controller computes

$$\Pi = \max(\rho, \sum_i c_i) + \sum_i g_i, \quad (2)$$

where ρ is the engine’s reported occupancy, c_i the context length of live session i , and g_i its forecast growth over the next horizon rounds. A session is admitted when $\Pi + n \leq B$, with n the candidate’s estimated need and B the budget. The three terms are respectively the measured physical occupancy, a reservation for sessions currently parked in a tool wait (whose KV is evictable and therefore absent from ρ), and a reservation for future growth.

D. Baselines and the verdict criterion

Baselines are static caps: at most N sessions resident, $N \in \{1, \dots, 5\}$, with and without HiCache. We compare on two axes: end-to-end wall clock (time from the first submission to the last completion) and SLO attainment (fraction of requests whose time-to-first-token is under 10 s).

The verdict criterion deserves care, because “faster than cap- N ” is not the question. The question is whether a policy reaches a point *above* the static front. A point that lies merely between two static caps is not a win: mixing cap- N and cap- M by time slice approximates any point on the line between them. So we take the upper-left envelope of the static points and report each controller’s signed distance above or below it. Every verdict in this paper is produced by one script from the raw logs, and the script’s own first version was wrong—it extrapolated past the end of the envelope and declared the slowest static cap a winner. We fixed it and report the episode because it is the failure mode this criterion is most prone to.

III. DECOMPOSING THE WALL CLOCK

A. Two components

We split the wall clock at every sampled instant according to whether the engine has any request in flight:

$$W = W_{\text{busy}} + W_{\text{idle}}. \quad (3)$$

Fig. 1 shows the split for every completed run. Two things are immediately visible and both matter later.

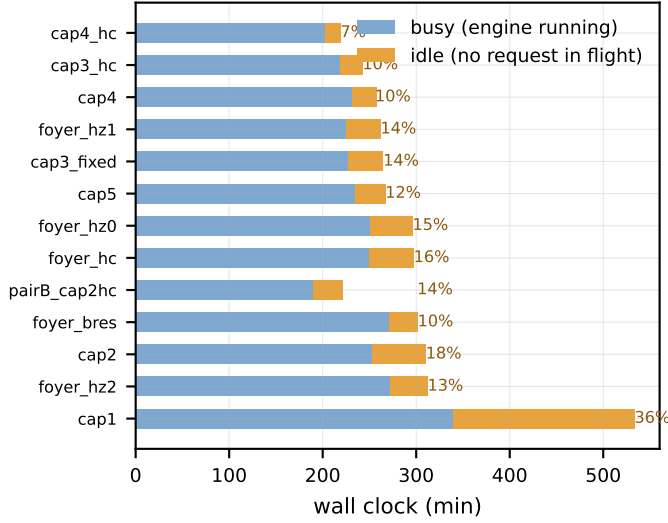


Fig. 1. Wall clock split into busy and idle for every completed run, ordered by total duration. Idle is 7.4% for the fastest configuration and 36.3% for the slowest. Note *bres*: it cuts idle to 10.1% and is still slower than its own baseline, because busy time grows instead.

First, idle time is not noise. It ranges from 7.4% of the wall clock for the fastest configuration to 36.3% for the slowest, and the stretches are long: median 5–10 s, p90 20–25 s, maximum 366 s. This is not a sampling artifact; we sampled every 5 s over runs of 3–9 hours.

Second, the two components move independently. *bres* (§IV) reduces idle from 15.9% to 10.1%, recovering about 17 minutes—and is nonetheless *slower* than its own baseline, because its busy time grows by 21 minutes. Any account that treats “more concurrency” as one thing will get this wrong.

B. What determines busy time

During busy time the engine executes steps; each step produces one token per running request. Hence

$$W_{\text{busy}} \approx \frac{G}{\bar{R}} \tau, \quad (4)$$

with G the total generated tokens (fixed by the trace at 706,010) and τ the per-step cost. Across our runs τ is not a fit parameter we tuned; we obtain it two ways and check that they agree.

a) *Method 1 (derived)*.: $\tau_1 = W \cdot \bar{R} / G$ using the engine’s own `num_running_reqs` for $\bar{R}$.

b) *Method 2 (independent)*.: For each step record we compute the per-request decode interval, $(\text{total_duration} - \text{first_token}) / (\text{output_len} - 1)$, and take the median over steps with `output_len` ≥ 32 .

The two disagree by a systematic 8–16%—Method 1 includes prefill and idle effects that Method 2 excludes—but they *order the runs identically* (Fig. 2). This matters because Method 1 alone would be circular: it derives τ from the quantity it is used to explain. Method 2 does not, and it recovers the same structure. We state the strength of this claim precisely: we have two consistent derived measurements of step time,

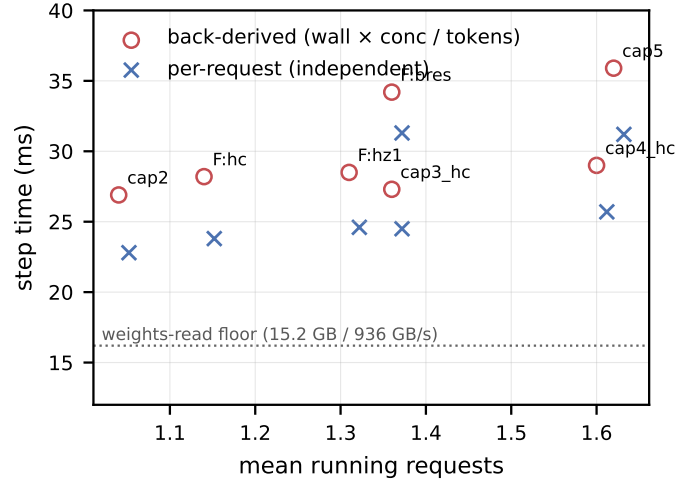


Fig. 2. Per-step cost by two independent measurements: circles derive it from the wall clock, crosses from per-request decode intervals. They differ by a systematic 8–16% but order the runs identically—*cap5* and *bres* are outliers in both. The dotted line is the cost of reading the weights once.

not an engine-reported one. The engine’s own latency counters (`itl_p50`, `e2e_p50`, `ttft_p50`) are exported as zero by SGLang v0.5.10 in our configuration, so no directly measured step time exists in our data; instrumenting this is future work.

C. The per-step cost is not constant in concurrency

Equation 4 would be uninteresting if τ were a constant. It is nearly so for most configurations—25.7–28.5 ms across seven runs—but two configurations are clear outliers, and both are ones that push occupancy hardest: *cap5* at 31.2 ms and *bres* at 31.3 ms, against 22.8–25.7 ms for the rest. Both outliers appear in both measurement methods, so this is not an artifact of either.

The mechanism is that a full pool forces evictions, and an evicted prefix must be recomputed when its session returns; that recomputation is chunked into decode steps, inflating each one. A back-of-envelope check supports this: reading the 15.2 GB of weights once at 936 GB/s costs 16.2 ms, so the observed 23–26 ms carries roughly 7–10 ms of non-decode work per step, and the outliers carry about 15 ms.

This is the crux of the paper. An admission policy can raise $\bar{R}$, and raising $\bar{R}$ is exactly what raises τ . The two effects partially cancel, and in our measurements they cancel almost completely.

D. Why idle time exists, and why it cannot be recovered

Idle time is a property of the workload, not of the controller. A session in a tool wait holds its permit and its KV but issues no request; with only about two live sessions, the probability that all of them are simultaneously waiting is substantial. We confirm the account two ways: idle fraction decreases monotonically with mean concurrency (Fig. 3), and the idle stretches are of the same order as the tool waits themselves.

The obvious remedy—keep more sessions live so that someone is always ready—is what *bres* does, and it fails

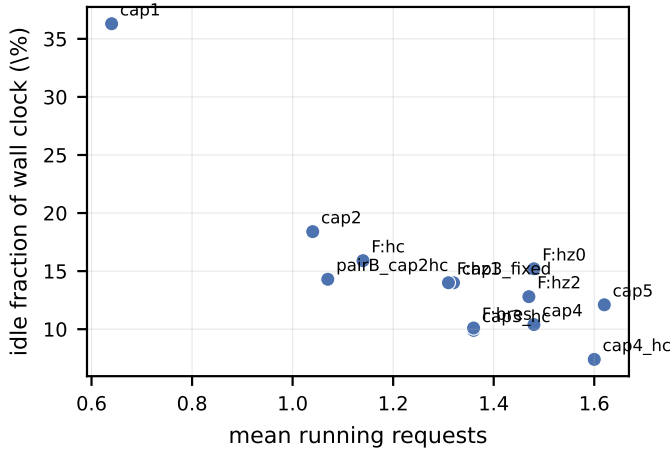


Fig. 3. Idle fraction falls monotonically with mean running requests. The cases that keep more sessions live waste less time waiting—and, per Fig. 2, pay for it in per-step cost.

for the reason given above. The structural remedy would be to make the engine do work during a wait, but there is no work to do: the session’s next round does not exist until its tool returns.

IV. RESULTS: FOUR ROUTES, ONE OUTCOME

A. The frontier

Fig. 4 places every completed run on the (wall clock, SLO) plane and draws the upper-left envelope of the static points. Table I gives the numbers and each controller’s signed distance from that envelope.

The result is unambiguous. Of nine controller variants, none is above the envelope. The best, a growth-reserve setting of one round instead of the default three, lands +0.20 percentage points above the line—inside the run-to-run spread of the same configuration, which we measured at 2.2–4.1%—and we therefore record it as *on* the line rather than above it.

Two features of the front are worth noting before the negative results. The envelope is steep between 213 and 236 minutes (0.76 pp per minute) and shallow thereafter (0.18 pp per minute), so the cost of buying SLO by retreating from the fast end is high. And W is not monotone in concurrency: five sessions (260.9 min) is *slower* than four (213.0 min), because the per-step cost rises faster than the request count does. That non-monotonicity is the first appearance of the mechanism that explains everything else.

B. Route 1: the accounting base

Our controller’s projection (Eq. 2) charges a session for its full context even while it is parked in a tool wait, when its KV is evictable and therefore absent from the engine’s reported occupancy. We argued this was a double charge and removed it, admitting on engine telemetry alone.

The charge is not double. Removing it did admit more—engine occupancy rose from 41.0% to 52.2% of the pool—and the round-1+ prefix hit rate held at 80.8% against 80.6%, so no eviction spiral occurred. But the run was 4.2 minutes *slower*

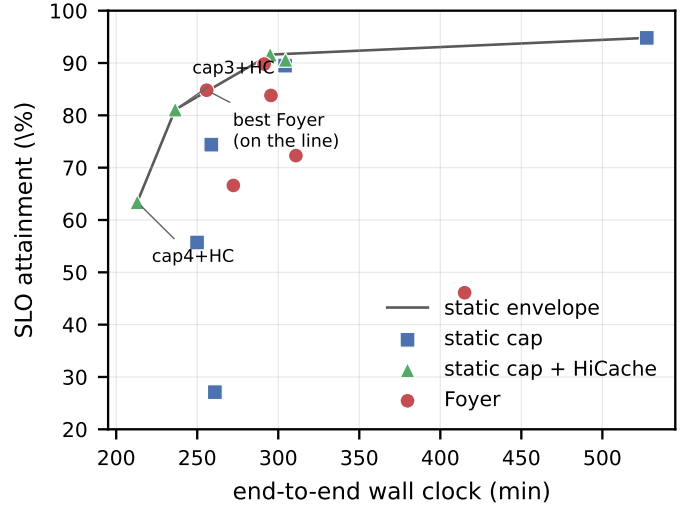


Fig. 4. Every completed run on the (wall clock, SLO) plane. The grey line is the upper-left envelope of the static points; a policy below or on it has not beaten a static cap, because time-slicing two caps approximates any point on the segment between them. Nine controller variants, none above.

TABLE I

ALL COMPLETED RUNS. Δ IS THE SIGNED DISTANCE FROM THE STATIC ENVELOPE AT THAT WALL CLOCK; POSITIVE WOULD MEAN A GENUINE WIN.

Policy	Wall (min)	Conc.	SLO (%)	Δ (pp)
<i>Static caps</i>				
cap1	527.4	1.00	94.8	—
cap2	304.2	1.99	89.5	−2.2
cap3 (fixed)	258.7	2.89	74.4	−10.7
cap4	249.9	3.94	55.7	−27.9
cap5	260.9	4.89	27.1	−58.4
<i>Static caps + HiCache</i>				
cap2+HC	295.0	1.99	91.6	<i>anchor</i>
cap3+HC	236.4	2.93	81.1	<i>anchor</i>
cap4+HC	213.0	3.93	63.4	<i>anchor</i>
<i>Foyer controller variants</i>				
horizon=1	255.8	2.58	84.8	+0.20
horizon=3 (base)	291.3	2.06	89.8	−1.14
horizon=2	306.1	2.22	86.9	−4.85
resident base	295.5	2.40	83.8	−7.82
horizon=0	290.0	2.69	77.2	−13.5
no gate, no resv.	272.3	3.25	66.6	−20.9
running gate=4	282.3	3.03	64.3	−24.0
all throttles off	415.0	2.58	46.1	−48.7

and lost 6.0 pp of SLO. Per §III, the extra occupancy raised both $\bar{R}$ and τ (31.3 ms against 23.8 ms), and the two cancelled.

We note that this claim had already been falsified once by source inspection: SGLang’s `token_usage` subtracts evictable blocks, so a parked session’s prefix is genuinely not counted, and charging for it reserves capacity that really will be needed back. The end-to-end measurement agrees with the source reading.

C. Route 2: the growth reserve

Sessions grow by roughly 2.5K tokens per round, so the controller reserves capacity for the next `horizon` rounds. We swept the horizon over $\{0, 1, 2, 3\}$ and the reserve is not a free parameter to be minimised: Fig. 5 shows the response is non-monotone. Removing the reserve entirely costs 13.5 pp of SLO, and a reserve of two rounds is 15 minutes *slower* than three. The optimum is at one round, and there it touches the static front and stops.

D. Route 3: the hard ceiling

Lowering the budget fraction is the crudest available knob and behaves as the other routes do: it slides the operating point along the front. We report it only for completeness, since it adds no mechanism.

E. Route 4: execution granularity

The first three routes all assume that what a controller can set is *how many sessions are resident*. A static cap sets the same quantity, so it is unsurprising that they land in the same place. The fourth route attacks the assumption instead: because a large fraction of live sessions are parked at any instant, a cap on *resident* sessions leaves the engine idle whenever all residents happen to be parked. Capping the *running-request* count instead should keep the engine busy while leaving the resident count high.

The permit design supports this directly: a paused session finishes its current round and blocks before its next one, and resumes when the controller re-admits it. We implemented it as a gate on the engine’s own `num_running_reqs` and swept the gate over $\{2, 4\}$.

It does not work, and the reason is structural rather than a tuning matter. Since a live session is running about 45% of the time, the running count is approximately $0.45 \times (\text{live count})$. Constraining the live count to N therefore constrains the running count to $0.45N$, which is below the gate—so the gate can only throttle, never pin. Both repairs fail: without a constraint on admission, a paused session is re-admitted on the next cycle and blocks for at most one 2s cycle; with one, the policy is a session cap again. The obstacle is granularity—*paused* takes effect at round boundaries, while pinning an instantaneous count requires control the permit mechanism does not have.

The measured outcome matches: the gate drove running concurrency from 1.42 to 1.29 and live sessions from 2.78 to 1.91. It tightened rather than pinned, and a tightening is another cap.

V. LOAD DEPENDENCE

A negative result about adaptivity invites the obvious objection: adaptivity pays when conditions *change*, and a single stationary trace is the one case where a static cap should be near-optimal. We agree, and we test it.

We run the same controller and the same cap family on a 25-session tier. The throttle configuration is reversed: with all safety throttles disabled, the 25-session tier runs *51% faster*

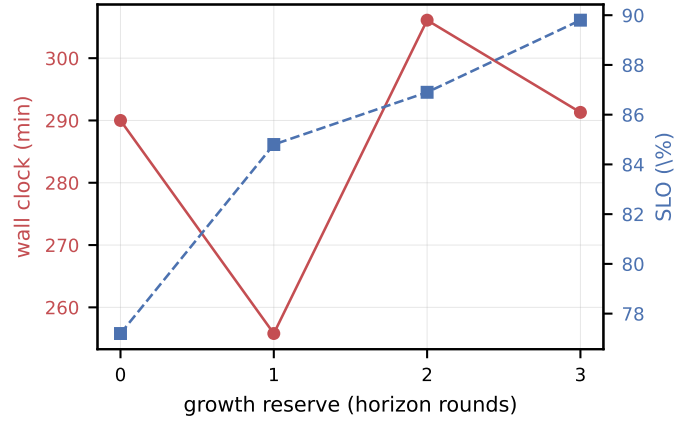


Fig. 5. The growth reserve is not a free parameter. Removing it costs 13.5 pp of SLO; two rounds is 15 minutes slower than three. The optimum at one round touches the static front and does not cross it.

(58.4 to 28.8 minutes) at a cost of only 2.4 pp of SLO, while on the 200-session tier disabling the same throttles costs 42% (291.3 to 415.0 minutes) and 43.7 pp of SLO. One switch, two regimes, opposite signs, both around 40–50% (Fig. 6). No single scalar setting is optimal across both.

This is the regime in which adaptive control should win, and we report two findings about it. The first is discouraging for our controller: on the 25-session tier it does not win. A static cap of three, re-run under the current harness, finishes in 24.2 minutes at 91.2% SLO—16% faster than the most aggressive Foyer configuration we measured on that tier, and with better run-to-run stability (segment SD 5.1 against 11.0 for cap-2). The stateless cap, which cannot see occupancy, backlog or remaining work, is simply better here. We flag one caveat and one gap. The caveat is that the tier comparison above is cross-generation: the 25-session throttle arms come from an earlier harness revision and are not directly comparable to the 200-session numbers, so we treat them as a direction rather than a magnitude. The gap is that the matching Foyer arm on the 25-session tier under the current harness was interrupted by another tenant and is not reported.

The second finding is the more useful one. The signal an adaptive controller would need is available—the number of sessions waiting for admission differs by 6–7 $\times$ between the tiers (median 9–11 at 25 sessions against 66 at 200, maximum 135)—but its *interpretation* is not obvious. At the 25-session tier the controller holds eleven sessions back while running exactly one; that backlog is manufactured by the throttling itself rather than evidence that demand has slackened. Reading it correctly requires combining instantaneous backlog with progress through the run, which is a stateful computation a cap cannot perform. We regard this as the most promising direction we did not finish, and we are explicit that we did not finish it.

VI. RELATED WORK

a) *Admission and scheduling for agentic serving.*: Recent systems report large gains from treating an agent

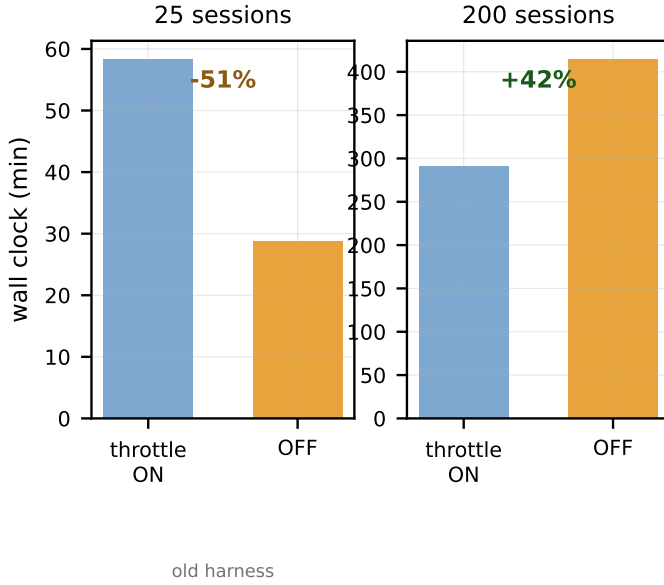


Fig. 6. The same throttle switch, two load tiers, opposite signs. The 25-session bars are from an earlier harness revision and are shown as a direction, not a magnitude; the matching current-harness arm was interrupted and is not reported.

session, rather than a request, as the schedulable unit, and from preserving session state across tool calls: session-aware admission with CPU-restorable KV [1], [2], program-level pause/resume with shortest-context-first restoration [3], and congestion-based control over an agent-level window [4]. A recurring measurement in this line is that recomputation, not admission, dominates: [4] attributes 49.1% of end-to-end latency to recomputation, and [5] preserves state on the CPU at the cost of a reload. Our results are consistent with this literature rather than in tension with it, and §VII explains why: the mechanism these systems add is one our controller already has.

b) Engine-side techniques.: A separate line improves the engine rather than the admission decision: chunked prefill with stall-free batching [6], prefill/decode disaggregation [7], and cache-centric scheduling [8]. Our §III attributes 7–10 ms of each 26 ms step to non-decode work, which is exactly what this line attacks. We emphasise that these techniques raise the achievable rate for *all* policies; they move the frontier rather than deciding who sits above it.

c) Prefix caching.: The prefix cache is often proposed as the signal an admission policy should use. In our workload it is not a policy variable: round-1+ hit rate equals the trace’s own planned value (89.8%) to within one percentage point across four different admission policies, so admission does not move it. We note the boundary of this claim—we varied admission policy, not eviction policy, and SGLang exposes alternate eviction policies we did not test.

VII. DISCUSSION

a) Why published gains do not reappear.: The systems cited above win by preserving session state across tool waits

and by pausing at agent granularity. Foyer inherits both properties from its permit design: a session holds its permit for its whole lifetime and its KV is preserved across waits, and the permit file supports the same pause primitive [4] uses. If that mechanism is where the published gains come from—and the measurements in those papers suggest it is—then a controller that already has it has nothing left to add by admitting more cleverly. This reframes our negative result: it is not that admission control cannot work, but that on this workload the mechanism that makes it work has already been spent.

b) A criterion, not just a verdict.: The decomposition suggests a cheap test for whether admission control can pay on a given workload. Measure $\bar{R}$ and τ (by Method 2, which needs only per-request logs) over a range of concurrency. If τ is flat, raising $\bar{R}$ converts directly into wall clock and admission control has room. If τ rises proportionally with $\bar{R}$, the two cancel and it does not. Our seven measured configurations give 22.8–31.3 ms across a running range of 1.04–1.62, i.e. a rise of about 8 ms over 0.58—enough to cancel the gain. We offer this as the paper’s constructive output, and we note it is testable in minutes on a system that already logs per-request timings.

c) Limitations.: Our results come from one 24 GB accelerator, one 7B model and one trace family; generality is untested. Two specific boundaries matter. First, the pool holds about four median sessions, so concurrency is capacity-bound by construction; on a device with a much larger pool the premise may not hold and the conclusion may invert—we regard reproducing on a larger device as the single most important next experiment. Second, the load is stationary, single-tenant and single-SLO. The load-dependence result of §V says the reverse should be expected when conditions vary, and multi-tenant or heterogeneous-SLO settings are where we would look next.

d) Reproducibility.: Every verdict in this paper is produced by one script from the raw per-step logs, and every run is guarded against a shrunken KV pool. We report three claims we retracted during the work, because the retractions were the most informative events: a median-based reading of the tool wait that inverted a central assumption, a growth-sensitivity conclusion that rested on an arm we had already invalidated, and the accounting double-charge of §IV, falsified twice—once by source inspection and once by the end-to-end measurement here.

VIII. CONCLUSION

We asked whether an adaptive admission controller can beat a tuned static concurrency cap on a memory-bound agentic LLM workload, and found that on our system it cannot: nine variants, four mechanistically distinct routes, none above the static front. The reason is a cancellation. Admission policy controls how many requests are in flight; the per-step cost rises with that number; and the product is what the wall clock sees. We gave a two-part decomposition of the wall clock that makes the cancellation measurable, identified a structural idle

component the policy cannot recover, and showed the regime in which adaptivity should have paid—where the same switch is worth +51% and −42%—is one where a stateless cap still wins. The constructive output is a per-request measurement that decides, in minutes, whether admission control has room on a given workload.

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